Fazoviy jismlarning yoyilmasi

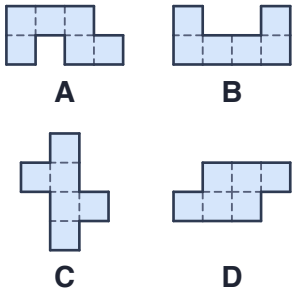
2-test (qiyinroq) · 20 ta savol

Ism-familiya: _____ Sinf: _____ Sana: _____

Har savolda bitta to'g'ri javobni belgilang.

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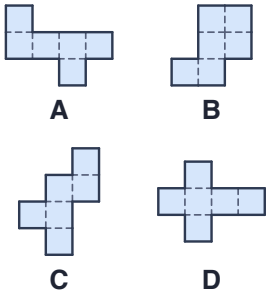
1. **Qaysi shakl kubning yoyilmasi?**



A) B
B) C
C) A
D) D

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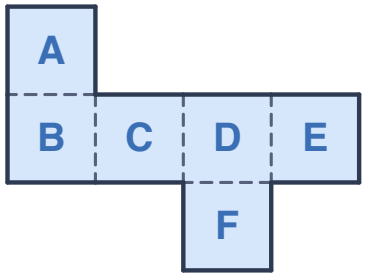
2. **Qaysi shakl kubning yoyilmasi EMAS?**



A) C
B) D
C) B
D) A

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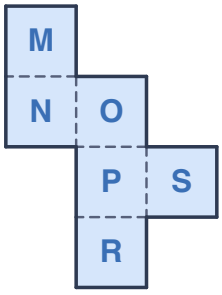
3. **Rasmdagi yoyilmadan kub yasaldi. A harfi qarshisida qaysi harf turadi?**



A) C
B) E
C) D
D) F

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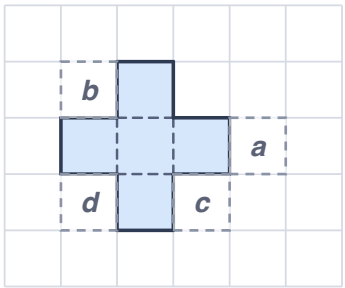
4. **Rasmdagi yoyilmadan kub yasaldi. Qaysi juftlik qarama-qarshi yoqlar EMAS?**



A) N va P
B) M va P
C) O va R
D) N va S

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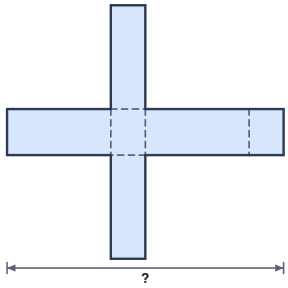
5. **Katakli qog'ozda 5 ta kvadrat bor. Kub yoyilmasi hosil bo'lishi uchun oltinchi kvadratni a, b, c, d joylardan qaysi biriga qo'yish kerak?**



A) c
B) b
C) d
D) a

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6. **Kuboidning bo'yi 9 cm, eni 4 cm, balandligi 3 cm. Rasmdagi yoyilmada «?» bilan belgilangan uzunlik necha cm?**

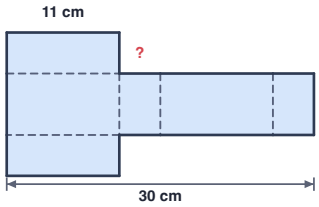


A) 26 cm **B)** 24 cm

C) 21 cm **D)** 16 cm

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7. **Rasmda kuboid yoyilmasi berilgan. «?» bilan belgilangan kesma necha cm?**



A) 19 cm **B)** 4 cm

C) 6 cm **D)** 8 cm

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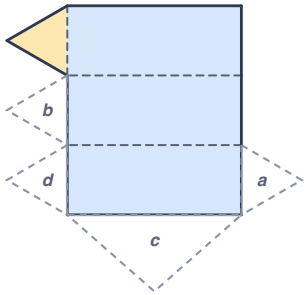
8. **Kuboid yoyilmasida 8 cm × 5 cm va 5 cm × 2 cm o'lchamli yoqlar bor. Uchinchi juft yoqlarning o'lchami qanday?**

A) 8 cm × 2 cm **B)** 8 cm × 5 cm

C) 2 cm × 2 cm **D)** 5 cm × 5 cm

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9. **Prizma yoyilmasiga yana bitta uchburchak kerak. a, b, c, d dan qaysi biri to'g'ri keladi?**

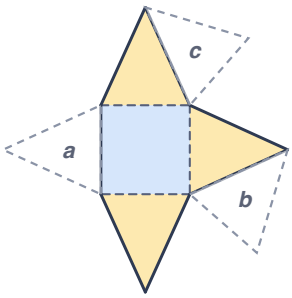


A) d **B)** c

C) b **D)** a

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10. **Asosi kvadrat bo'lgan piramidaning yoyilmasida bitta uchburchak yetishmaydi. a, b, c dan qaysi biri to'g'ri keladi?**



A) Hech biri **B)** a

C) c **D)** b

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11. **Fazoviy jismning yoqlari 5 ta, uchlari 6 ta. Y + U − Q = 2 ekani ma'lum. Qirralari soni nechta?**

A) 11 **B)** 13 **C)** 7 **D)** 9

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12. **Piramidaning yoyilmasi 1 ta kvadrat va 4 ta uchburchakdan iborat. Piramidaning nechta uchi va nechta qirrasi bor?**

A) 5 ta uch, 8 ta qirra **B)** 4 ta uch, 8 ta qirra

C) 8 ta uch, 5 ta qirra **D)** 5 ta uch, 4 ta qirra

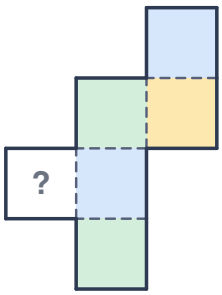
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13. **Kuboid yoyilmasida 2 ta kvadrat va 4 ta teng to'g'ri to'rtburchak bor. Kvadratning tomoni 3 cm, to'g'ri to'rtburchakning uzun tomoni 10 cm. Kuboidning o'lchamlari qanday?**

A) 10 cm, 3 cm, 6 cm
B) 10 cm, 3 cm, 3 cm
C) 3 cm, 3 cm, 3 cm
D) 10 cm, 10 cm, 3 cm

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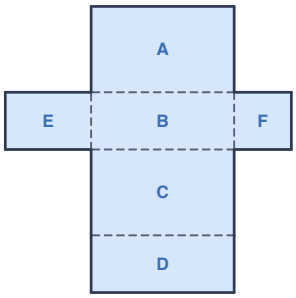
14. **Kubning qarama-qarshi yoqlari bir xil rangga bo'yalishi kerak. Yoyilmadagi «?» kvadrat qaysi rangda bo'ladi?**



A) ko'k
B) yashil
C) sariq
D) qizil

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15. **Kuboidning bitta yog'i yoyilmada noto'g'ri chizilgan. Qaysi biri?**



A) C
B) F
C) E
D) D

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16. **Kub yoyilmasida 5 ta kvadrat bitta qatorda turishi mumkinmi?**

A) Ha, 6-kvadrat istalgan joyda bo'lsa
B) Yo'q — 5-kvadrat 1-kvadrat bilan bitta yoqqa tushadi
C) Yo'q — kubda 5 ta yoq yo'q
D) Ha, 6-kvadrat o'rtadagisiga yopishsa

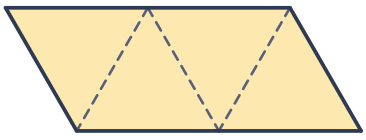
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17. **Uchburchakli prizma yoyilmasidagi to'g'ri to'rtburchaklar 4 cm × 9 cm. Prizmaning asoslari teng tomonli uchburchaklar. Uchburchakning tomoni necha cm?**

A) 9 cm
B) 13 cm
C) 36 cm
D) 4 cm

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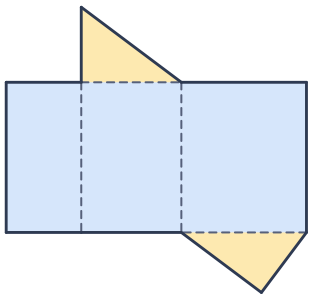
18. **Rasmdagi yoyilmadan hosil bo'lgan jismning nechta uchi bor?**



A) 8
B) 4
C) 6
D) 3

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19. **Rasmdagi yoyilmadan qanday jism hosil bo'ladi?**



A) Asosi uchburchak bo'lgan piramida
B) Kuboid
C) Bu yoyilmadan jism hosil bo'lmaydi
D) Uchburchakli prizma

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20. Ikki xil yoyilma bitta kuboidniki bo'lishi uchun ularda nima bir xil bo'lishi kerak?

A) Yoyilmaning rangi

B) Yoyilmaning umumiy uzunligi

C) Yoqlarning joylashuvi

D) Yoqlar (to'g'ri to'rtburchaklar) o'lchamlari

Javoblar

Fazoviy jismlarning yoyilmasi · 2-test (qiyinroq)

№	Javob	Qiymati
1	**B**	C
2	**C**	B
3	**D**	F
4	**A**	N va P
5	**D**	a
6	**B**	24 cm
7	**B**	4 cm
8	**A**	8 cm × 2 cm
9	**D**	a
10	**B**	a

№	Javob	Qiymati
11	**D**	9
12	**A**	5 ta uch, 8 ta qirra
13	**B**	10 cm, 3 cm, 3 cm
14	**C**	sariq
15	**B**	F
16	**B**	Yo'q — 5-kvadrat 1-kvadrat bilan bitta yoqqa tushadi
17	**D**	4 cm
18	**B**	4
19	**D**	Uchburchakli prizma
20	**D**	Yoqlar (to'g'ri to'rtburchaklar) o'lchamlari

Qisqa yechimlar

1. A — yuqori qatorning o'rtasini tub desak, chap va o'ng ustunlarning pastki kvadratlari ikkalasi ham old devorga tushadi.
B — qatordagi 4 ta kvadratning bir tomonida ikkita kvadrat: ikkalasi orqa devorga tushadi.
D — o'rtadagi 4 ta kvadrat «deraza» (2 × 2).
C — ustundagi 4 ta kvadratning chap va o'ng tomonida bittadan kvadrat: kub yoyilmasi.

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2. A, C, D ni xayolan buklasak, har kvadrat alohida yoqqa tushadi.
B: o'rta ustunning pastki kvadratini tub desak, uning ustidagi — orqa devor, eng tepadagi — qopqoq.
O'ngdagi ikki kvadratning pastkisi — o'ng devor, tepadagisi esa yana qopqoqqa tushadi — ustma-ust (o'ngdagi to'rtta kvadrat «deraza» bo'lib turibdi). Javob: **B**.

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3. B, C, D, E — bir qatorda: B va D, C va E qarama-qarshi (ketma-ket uchtaning chetlari).
Qolgan ikki yoq — A va F — bir-biriga qarama-qarshi.
Javob: **F**.

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4. P ni tub deymiz: O — orqa, R — old, S — o'ng devor, N — chap devor, M — qopqoq.
Qarama-qarshi juftlar: P va M, O va R, N va S.
N (chap devor) va P (tub) — qo'shni yoqlar. Javob: **N va P**.

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5. O'rtadagi kvadrat — tub, uning to'rt tomonidagilar — 4 ta devor. Faqat qopqoq yetishmaydi.
b, c, d burchakda — ikkita devorga tegib turadi, buklanganda devorlardan biriga tushadi.
a o'ng devorning davomida turibdi: o'ng devor tik turganda u tepaga buklanib, qopqoq bo'ladi. Javob: **a**.

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6. Qatorda 4 ta yoq turibdi: ikkitasi 9 cm × 4 cm (keng), ikkitasi 3 cm × 4 cm (tor).
Qator uzunligi: 9 + 3 + 9 + 3 = **24 cm**.

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7. Qatordagi 4 ta yoq: ikkitasi 11 cm, ikkitasi teng noma'lum.
30 − 11 − 11 = 8 cm — ikkita tor yoq birga.
Bittasi: 8 : 2 = **4 cm**.

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8. Kuboidning uchta o'lchami: 8 cm, 5 cm, 2 cm.
Har yoq ikkita o'lchamdan iborat: 8 × 5, 5 × 2 bor — yetishmaydigani 8 va 2.
Javob: **8 cm × 2 cm**.

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9. Chapdagi uchburchak «quvur»ning chap uchini yopadi. Ikkinchisi o'ng uchini yopishi kerak.
b va d — yana chap tomonda; c — to'rtburchakning uzun tomoniga yopishgan, u yerda uch emas.
Javob: **a** — o'ng uchda.

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10. Tepa, o'ng va past yon yoqlar bor, chap yon yoq yo'q.
b — o'ng uchburchakka pastdan yopishgan: buklanganda pastdagi yon yoq o'rniga tushadi. c — tepadagi uchburchakka yopishgan: o'ng yon yoq o'rniga tushadi.
a kvadratning bo'sh chap tomonida — chap yon yoq. Javob: **a**.

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11. 5 + 6 − Q = 2.
11 − Q = 2, demak Q = 11 − 2 = **9**.
(Bu — uchburchakli prizma.)

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12. Uchlar: kvadratning 4 ta uchi + piramidaning uchi = 5.
Qirralar: asosda 4 ta + uchdan asosga tushgan 4 ta = 8.
Javob: **5 ta uch, 8 ta qirra**.

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13. Kvadrat yoq — kuboidning eni va balandligi: 3 cm va 3 cm.
To'g'ri to'rtburchakning qisqa tomoni kvadratga tutashadi — u ham 3 cm, uzun tomoni 10 cm — bo'yi.
Javob: **10 cm, 3 cm, 3 cm**.

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14. Ustundagi uchta kvadrat (yashil, ko'k, yashil): chetdagilari qarama-qarshi — ikkalasi yashil.
Tepadagi ko'k va ustundagi ko'k ham qarama-qarshi.
Qolgan ikki yoq — sariq va «?» — qarama-qarshi. Javob: **sariq**.

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15. E va F — kuboidning yon yoqlari, ular teng bo'lishi kerak.
E ning eni A ning qisqa tomoniga teng, F esa torroq.
F buklanganda A va C ga yetmaydi. Javob: **F**.

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16. Qatordagi kvadratlar tub, devor, qopqoq, devor bo'lib aylanadi — 4 tasi kubni to'liq aylanib chiqadi.
5-kvadrat yana 1-kvadrat turgan joyga tushadi.
Javob: **yo'q**.

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17. Uchburchak to'g'ri to'rtburchakning qisqa tomoniga tutashadi (prizma uchida).
Tutashadigan tomonlar teng: **4 cm**.
9 cm — prizmaning uzunligi (yon qirrasi).

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18. 4 ta teng uchburchak — asosi uchburchak bo'lgan piramida.
Uchlari: asosning 3 ta uchi + piramidaning uchi = **4**.

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19. 2 ta teng uchburchak va 3 ta to'g'ri to'rtburchak bor. To'rtburchaklarning eni 3, 4, 5 — uchburchak tomonlariga teng.
To'rtburchaklar har xil bo'lsa ham, ular uchburchak tomonlariga mos: buklanadi.
Bu — **uchburchakli prizma**.

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20. Bitta kuboidning yoyilmalarida yoqlar har xil joylashishi mumkin, umumiy uzunligi ham o'zgaradi.
Lekin yoqlar o'sha 6 ta: 3 juft teng to'g'ri to'rtburchak.
Javob: **yoqlar o'lchamlari**.
